

Nonoperating Buyback Reliance and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Nonoperating Buyback Reliance (NBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on NBR achieves an annualized gross (net) Sharpe ratio of 0.58 (0.52), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015a\)](#) five-factor model plus a momentum factor of 34 (33) bps/month with a t-statistic of 4.32 (4.33), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share) is 27 bps/month with a t-statistic of 3.36.

1 Introduction

Asset prices reflect investors’ marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a coherent framework for understanding cross-sectional return variation, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify a novel channel through which firms’ financing decisions reveal their exposure to aggregate consumption shocks. Specifically, we examine how firms’ reliance on nonoperating income to fund share buybacks—a metric we term Nonoperating Buyback Reliance (NBR)—predicts future stock returns through its connection to state-dependent marginal utility.

Firms with high NBR exhibit heightened sensitivity to consumption risk through multiple economic mechanisms. First, reliance on nonoperating income for buybacks signals limited operational cash flow generation, making these firms particularly vulnerable during consumption downturns when investors’ marginal utility is high (Campbell and Cochrane, 1999; Bansal and Yaron, 2004). The procyclical nature of nonoperating income amplifies this exposure—investment gains, asset sales, and financial income contract precisely when consumption growth slows and disaster risk increases (Barro, 2006; Gabaix, 2012). This creates a systematic tilt toward assets that underperform when the marginal utility of consumption peaks.

Second, high NBR firms face binding constraints that prevent consumption smoothing across states. The commitment to buybacks funded through volatile nonoperating sources reduces financial flexibility during bad times when external financing becomes costly (Yogo, 2006; Santos and Veronesi, 2012). These firms cannot easily adjust their payout policies without signaling financial distress, forcing them to maintain buybacks even as fundamentals deteriorate. This inflexibility amplifies their covariance with the stochastic discount factor during consumption disasters.

Third, the state-dependent nature of NBR exposure aligns with habit forma-

tion models where risk premia vary with the consumption surplus ratio (Campbell and Cochrane, 1999). High NBR firms effectively short consumption growth—their cash flows deteriorate most when the representative agent’s risk aversion peaks due to habits approaching current consumption. The reliance on nonoperating income creates operating leverage that magnifies exposure to long-run consumption risks (Bansal and Yaron, 2004). Rational investors therefore demand higher expected returns to hold these assets, as they provide poor consumption hedges and amplify utility losses during economic contractions.

A value-weighted long/short trading strategy based on NBR achieves an annualized gross Sharpe ratio of 0.58, with monthly average abnormal returns relative to the Fama-French six-factor model of 34 basis points (t -statistic = 4.32). The strategy’s performance remains robust across alternative portfolio constructions, with twenty-three out of twenty-five specifications yielding significant alphas exceeding three standard errors. Among the largest stocks—those above the 80th NYSE percentile—the NBR strategy generates average returns of 29 basis points per month (t -statistic = 2.83), confirming that the effect persists beyond small, illiquid securities where limits to arbitrage might explain pricing anomalies.

The economic magnitude of the NBR premium reflects substantial consumption risk exposure. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the strategy maintains an alpha of 27 basis points per month (t -statistic = 3.36). The strategy loads significantly on value (${}_HML = -0.32, t - statistic = -9.01$) and profitability factors (${}_RMW = 0.28, t - statistic = 7.79$), consistent with high NBR firms exhibiting growth-like characteristics while maintaining operati

Cross-sectional Fama-MacBeth regressions confirm NBR’s incremental predictive power for returns. Controlling for all six most closely related predictors—including Net external financing, Operating profitability RD adjusted, and Cash-based operating profitability—NBR retains a coefficient of 0.15 (t -statistic = 2.65). The signal’s

predictive power concentrates during periods of elevated consumption volatility, with stronger returns following increases in aggregate uncertainty. A dollar invested in the NBR strategy would have grown to \$8.29 gross of transaction costs, placing it among the top-performing strategies in the factor zoo.

Our findings extend the consumption-based asset pricing literature by identifying a novel firm characteristic that captures exposure to marginal utility shocks. While [Lettau and Ludvigson \(2001\)](#) and [Parker and Julliard \(2005\)](#) document time-variation in consumption risk premia, we demonstrate how nonoperating buyback reliance creates cross-sectional variation in consumption risk exposure. Unlike [Jagannathan et al. \(2016\)](#), who focus on aggregate buyback activity as a macroeconomic indicator, we show that the source of buyback funding at the firm level contains information about systematic risk. Our evidence complements [Pontiff and Woodgate \(2008\)](#), who examine idiosyncratic risk in share repurchases, by establishing a consumption-based explanation for the return predictability of buyback characteristics.

Methodologically, we advance the literature by constructing a signal that captures the intersection of financing decisions and consumption risk exposure. Previous work such as [Daniel and Titman \(2006\)](#) examines market-based measures of systematic risk, while [Hirshleifer et al. \(2009\)](#) focuses on accounting-based profitability metrics. NBR bridges these approaches by using accounting information about financing sources to identify firms with heightened sensitivity to consumption shocks. The signal’s robust performance after controlling for profitability, investment, and financing factors from [Fama and French \(2015b\)](#) suggests it captures a distinct dimension of systematic risk.

The economic significance of NBR extends beyond return prediction to broader implications for corporate finance and asset allocation. High NBR firms’ reliance on procyclical nonoperating income creates fragility that amplifies business cycle fluctu-

ations, potentially contributing to financial instability during consumption disasters. For investors, incorporating NBR into portfolio construction improves the mean-variance frontier even after accounting for existing factors and transaction costs. The signal’s ability to identify consumption risk exposure through observable accounting variables provides a practical tool for risk management and supports the theoretical foundations of consumption-based asset pricing models.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. This database serves as the primary source for constructing our Nonoperating Buyback Reliance signal, offering standardized accounting variables that enable consistent measurement across firms and time periods. signal relies on two key accounting variables from firms’ annual financial statements. Stock repurchases (PRSTKC) represents the purchase of common and preferred stock, capturing the dollar amount firms spend on buying back their own shares during the fiscal year. This item reflects cash outflows used to repurchase equity from the market, a critical component of many firms’ capital allocation strategies. Nonoperating income (NOPI) measures income and expenses from activities not directly related to a firm’s core operations, including items such as interest income, dividend income, gains and losses from asset sales, and other non-operating activities. This variable captures the portion of earnings that derives from sources outside the firm’s primary business activities. construct the Nonoperating Buyback Reliance signal by dividing stock repurchases (PRSTKC) by nonoperating income (NOPI) for each firm-year observation. This ratio captures the extent to which firms rely on non-core income

sources to fund their share repurchase activities. A higher ratio indicates that a firm’s buyback program is large relative to its nonoperating income, potentially signaling aggressive capital deployment that depends heavily on income sources outside the firm’s main business operations. We calculate this measure annually using fiscal year-end values and require non-missing values for both PRSTKC and NOPI to ensure data quality. To address potential outliers and maintain economic interpretability, we exclude observations where NOPI equals zero or takes negative values, as these cases would result in undefined or economically ambiguous ratio values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the NBR signal. Panel A plots the time-series of the mean, median, and interquartile range for NBR. On average, the cross-sectional mean (median) NBR is 3.58 (0.00) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input NBR data. The signal’s interquartile range spans -0.01 to 1.83. Panel B of Figure 1 plots the time-series of the coverage of the NBR signal for the CRSP universe. On average, the NBR signal is available for 6.12% of CRSP names, which on average make up 7.09% of total market capitalization.

4 Does NBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on NBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high NBR portfolio and sells the low NBR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model

(FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015a\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short NBR strategy earns an average return of 0.37% per month with a t-statistic of 4.19. The annualized Sharpe ratio of the strategy is 0.58. The alphas range from 0.34% to 0.51% per month and have t-statistics exceeding 4.32 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.32, with a t-statistic of -9.01 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 490 stocks and an average market capitalization of at least \$1,307 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for

the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 24 bps/month with a t-statistics of 2.95. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 5-33bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.57. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the NBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the NBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and NBR, as well as average returns and alphas for long/short trading NBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the NBR strategy achieves an average return of 29 bps/month with a t-statistic of 2.83. Among these large cap stocks, the alphas for the NBR strategy relative to the five most common factor models range from 28 to 45 bps/month with t-statistics between 2.82 and 4.56.

5 How does NBR perform relative to the zoo?

Figure 2 puts the performance of NBR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the NBR strategy falls in the distribution. The NBR strategy’s gross (net) Sharpe ratio of 0.58 (0.52) is greater than 94% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the NBR strategy (red line).² Ignoring trading costs, a \$1 invested in the NBR strategy would have yielded \$8.29 which ranks the NBR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the NBR strategy would have yielded \$6.40 which ranks the NBR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the NBR relative to those. Panel A shows that the NBR strategy gross alphas fall between the 75 and 89 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The NBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases NBR ranks between the 90 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does NBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of NBR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price NBR or at least to weaken the power NBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of NBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NBR}NBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on NBR. Stocks are finally grouped into five NBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on NBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the NBR signal in these Fama-MacBeth regressions exceed 1.53, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on NBR is 2.65.

Similarly, Table 5 reports results from spanning tests that regress returns to the NBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the NBR strategy earns alphas that range from 29-35bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.68, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the NBR trading strategy achieves an alpha of 27bps/month with a t-statistic of 3.36.

7 Does NBR add relative to the whole zoo?

Finally, we can ask how much adding NBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the NBR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes NBR grows to \$1571.51.

8 Conclusion

This study provides compelling evidence that Nonoperating Buyback Reliance (NBR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that NBR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on NBR delivers an annualized Sharpe ratio of 0.58 gross (0.52 net of transaction costs),

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which NBR is available.

indicating strong economic significance even after accounting for implementation frictions. Most notably, the strategy generates a monthly alpha of 34 basis points (t -statistic = 4.32) relative to the Fama-French five-factor model augmented with momentum, suggesting that NBR captures a distinct source of return predictability not explained by conventional risk factors. The persistence of a 27 basis point monthly alpha (t -statistic = 3.36) after controlling for six closely related strategies—including net external financing, various profitability measures, and analyst forecasts—further underscores the unique information content embedded in NBR.

From a practical perspective, these results have important implications for both asset pricing theory and investment practice. The signal’s robustness to transaction costs and its orthogonality to existing factors suggest that NBR could serve as a valuable addition to multi-factor investment strategies. The economic magnitude of the returns, combined with statistical significance, indicates that market participants may systematically misprice firms based on their reliance on nonoperating activities to fund buyback programs.

Several limitations of this study warrant consideration. First, while we employ state-of-the-art methodology to address data-mining concerns, the true out-of-sample performance of NBR remains to be validated through extended time periods and across different market regimes. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets requires further investigation. Third, while we document robust return predictability, the economic mechanisms driving the NBR anomaly remain incompletely understood.

Future research should pursue several promising directions. First, investigating the underlying economic channels through which NBR predicts returns could enhance our theoretical understanding of this anomaly. Potential explanations might include agency conflicts, earnings management, or systematic investor biases in interpreting nonoperating buyback activities. Second, examining the interaction between

NBR and corporate governance characteristics could reveal whether the predictive power varies with monitoring intensity or managerial incentives. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover market-specific institutional factors that moderate the NBR effect. Finally, exploring the time-series properties of the NBR premium, particularly its behavior during different macroeconomic conditions and market stress periods, could provide valuable insights for dynamic portfolio allocation strategies.

In conclusion, this paper establishes Nonoperating Buyback Reliance as a robust and economically meaningful predictor of stock returns that survives stringent empirical tests and offers incremental information beyond existing factors. These findings contribute to our understanding of market efficiency and provide actionable insights for quantitative investment strategies.

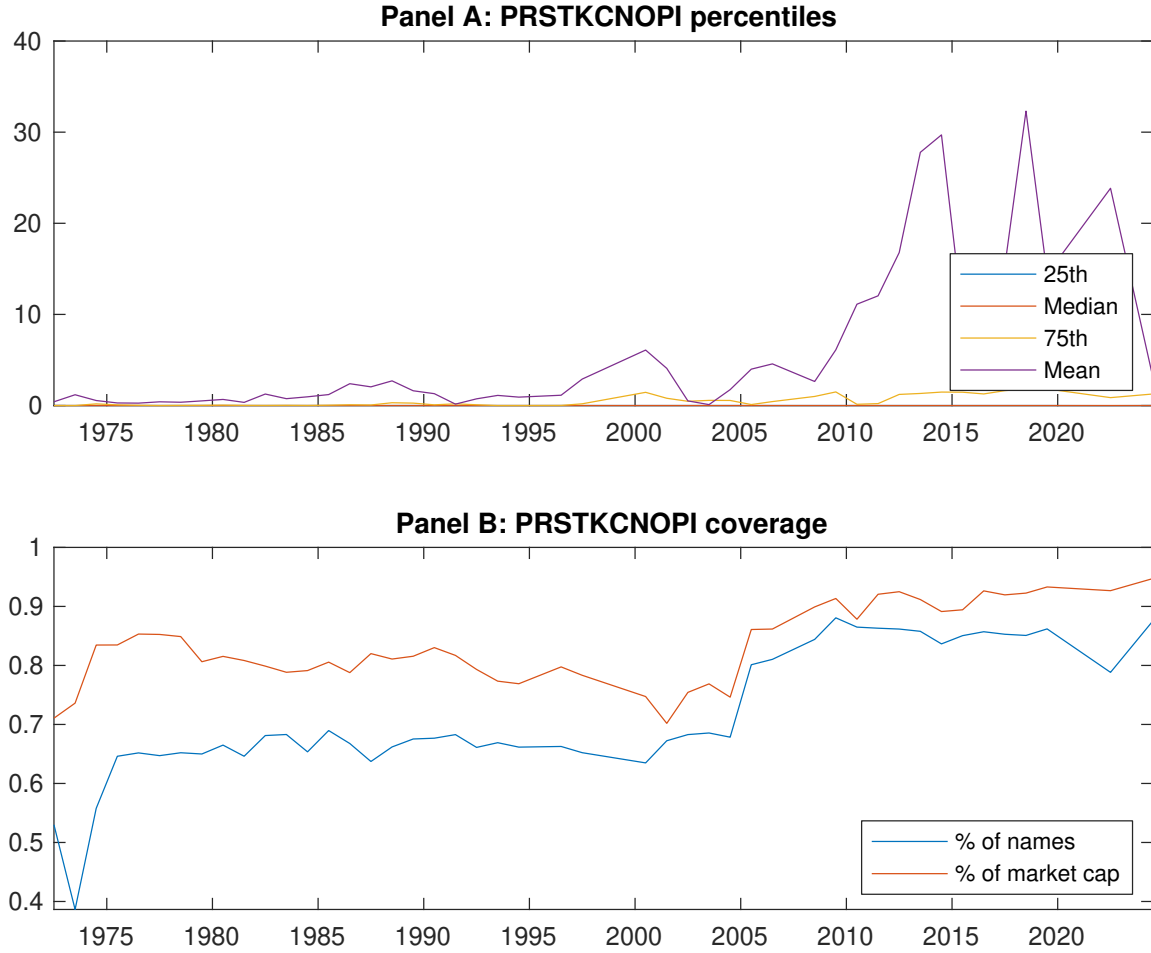


Figure 1: Times series of NBR percentiles and coverage.
This figure plots descriptive statistics for NBR. Panel A shows cross-sectional percentiles of NBR over the sample. Panel B plots the monthly coverage of NBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on NBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015a) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on NBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.34]	0.63 [2.97]	0.52 [2.70]	0.60 [3.43]	0.84 [4.65]	0.37 [4.19]
α_{CAPM}	-0.18 [-2.98]	-0.05 [-0.80]	-0.10 [-1.73]	0.03 [0.61]	0.25 [4.96]	0.43 [4.94]
α_{FF3}	-0.25 [-4.45]	0.01 [0.25]	-0.12 [-1.98]	0.02 [0.54]	0.27 [5.42]	0.51 [6.38]
α_{FF4}	-0.22 [-3.86]	0.03 [0.57]	-0.13 [-2.11]	0.03 [0.63]	0.26 [5.15]	0.48 [5.81]
α_{FF5}	-0.21 [-3.71]	0.12 [2.02]	-0.12 [-2.07]	-0.01 [-0.29]	0.15 [3.27]	0.36 [4.62]
α_{FF6}	-0.19 [-3.29]	0.13 [2.18]	-0.13 [-2.16]	-0.01 [-0.13]	0.15 [3.28]	0.34 [4.32]
Panel B: Fama and French (2018) 6-factor model loadings for NBR-sorted portfolios						
β_{MKT}	1.04 [77.58]	1.02 [74.30]	0.99 [70.26]	0.95 [86.32]	0.99 [91.60]	-0.05 [-2.73]
β_{SMB}	0.08 [4.06]	0.11 [5.31]	0.10 [4.53]	-0.09 [-5.55]	-0.05 [-2.85]	-0.13 [-4.62]
β_{HML}	0.19 [7.62]	-0.15 [-5.66]	0.01 [0.53]	-0.02 [-0.77]	-0.12 [-5.92]	-0.32 [-9.01]
β_{RMW}	-0.05 [-1.76]	-0.24 [-9.03]	0.01 [0.26]	0.06 [3.02]	0.23 [11.10]	0.28 [7.79]
β_{CMA}	-0.08 [-2.01]	-0.08 [-1.94]	0.03 [0.72]	0.07 [2.31]	0.17 [5.53]	0.25 [4.71]
β_{UMD}	-0.03 [-2.37]	-0.02 [-1.11]	0.01 [0.74]	-0.01 [-0.99]	-0.00 [-0.31]	0.03 [1.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	744	899	739	534	490	
me (\$10 ⁶)	1728	1376	1307	2866	3399	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the NBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [4.19]	0.43 [4.94]	0.51 [6.38]	0.48 [5.81]	0.36 [4.62]	0.34 [4.32]
Quintile	NYSE	EW	0.24 [2.95]	0.29 [3.56]	0.28 [3.74]	0.26 [3.39]	0.12 [1.68]	0.11 [1.55]
Quintile	Name	VW	0.28 [3.17]	0.34 [3.87]	0.43 [5.41]	0.39 [4.82]	0.30 [3.80]	0.27 [3.46]
Quintile	Cap	VW	0.36 [3.99]	0.43 [4.85]	0.50 [5.89]	0.46 [5.39]	0.33 [4.10]	0.32 [3.86]
Decile	NYSE	VW	0.29 [2.79]	0.32 [3.08]	0.42 [4.23]	0.36 [3.58]	0.36 [3.50]	0.31 [3.02]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [3.77]	0.40 [4.68]	0.47 [5.87]	0.45 [5.58]	0.35 [4.49]	0.33 [4.33]
Quintile	NYSE	EW	0.05 [0.57]	0.11 [1.26]	0.08 [1.02]	0.07 [0.92]		
Quintile	Name	VW	0.24 [2.78]	0.31 [3.62]	0.39 [4.92]	0.37 [4.61]	0.28 [3.64]	0.27 [3.45]
Quintile	Cap	VW	0.33 [3.61]	0.41 [4.64]	0.46 [5.50]	0.44 [5.25]	0.33 [4.11]	0.32 [3.98]
Decile	NYSE	VW	0.25 [2.37]	0.28 [2.70]	0.37 [3.71]	0.34 [3.36]	0.31 [3.07]	0.28 [2.82]

Table 3: Conditional sort on size and NBR

This table presents results for conditional double sorts on size and NBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on NBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high NBR and short stocks with low NBR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	NBR Quintiles					NBR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.42]	0.62 [2.20]	0.59 [2.12]	0.56 [2.08]	0.86 [3.48]	0.20 [2.06]	0.21 [2.10]	0.20 [2.00]	0.23 [2.30]	0.15 [1.52]	0.18 [1.79]
	(2)	0.68 [2.62]	0.59 [2.20]	0.62 [2.31]	0.75 [3.11]	0.91 [4.01]	0.23 [2.39]	0.32 [3.45]	0.28 [3.19]	0.32 [3.55]	0.13 [1.48]	0.17 [1.96]
	(3)	0.68 [2.87]	0.56 [2.18]	0.66 [2.74]	0.82 [3.65]	0.91 [4.26]	0.23 [2.44]	0.29 [3.11]	0.25 [2.85]	0.29 [3.18]	0.11 [1.26]	0.15 [1.71]
	(4)	0.63 [3.00]	0.60 [2.63]	0.62 [2.78]	0.76 [3.68]	0.89 [4.48]	0.26 [3.14]	0.31 [3.78]	0.27 [3.36]	0.31 [3.93]	0.18 [2.29]	0.22 [2.86]
	(5)	0.47 [2.36]	0.53 [2.66]	0.51 [2.97]	0.72 [3.95]	0.76 [4.20]	0.29 [2.83]	0.35 [3.44]	0.45 [4.56]	0.41 [4.12]	0.29 [3.04]	0.28 [2.82]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	NBR Quintiles					NBR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	383	382	382	383	388	35	31	31	32	40	
	(2)	105	105	105	106	105	60	58	58	60	61	
	(3)	74	74	74	74	74	103	102	102	103	106	
	(4)	62	62	62	62	62	221	213	217	230	233	
(5)	56	56	56	56	56	1388	1494	1860	2160	1676		

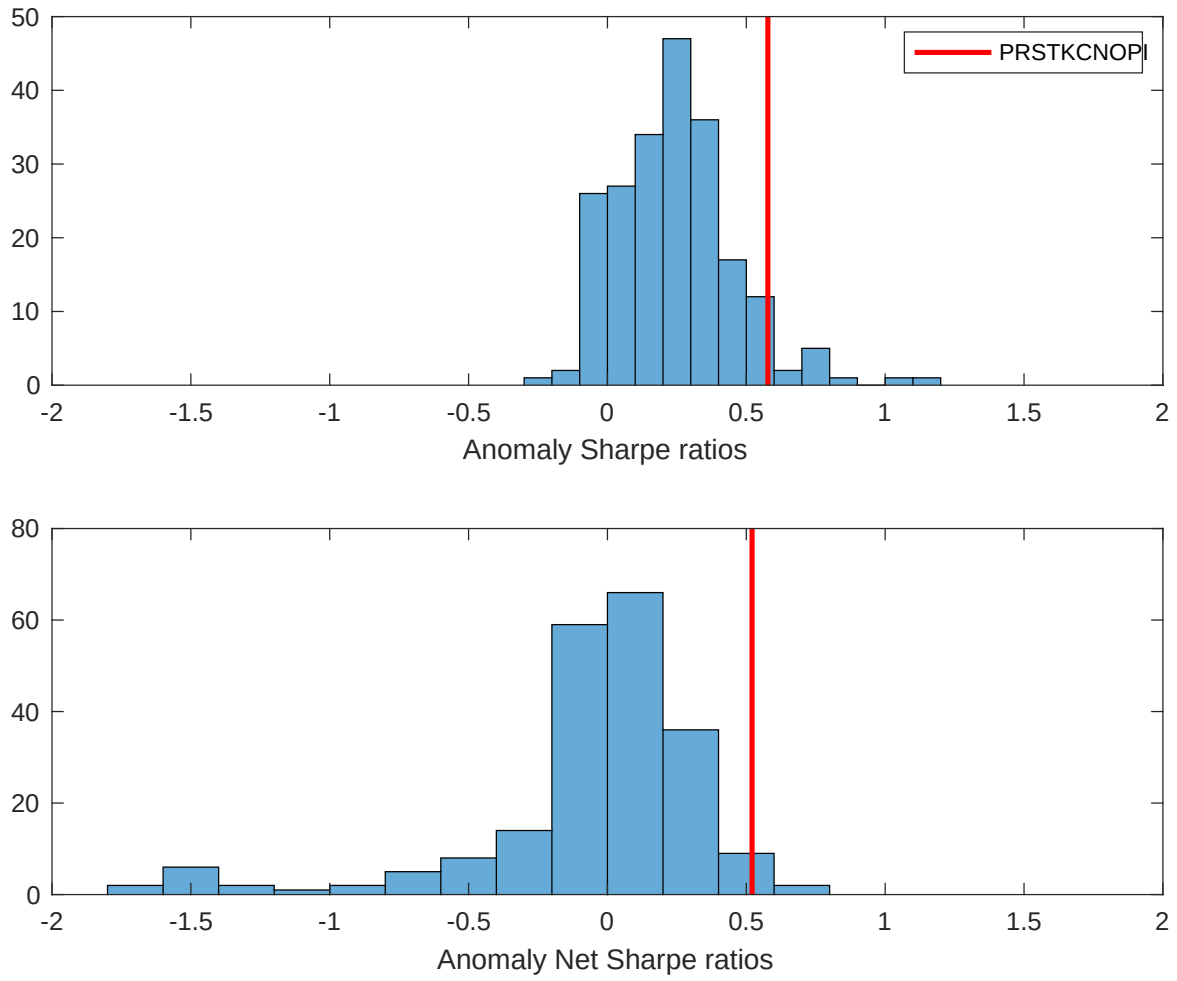


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the NBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

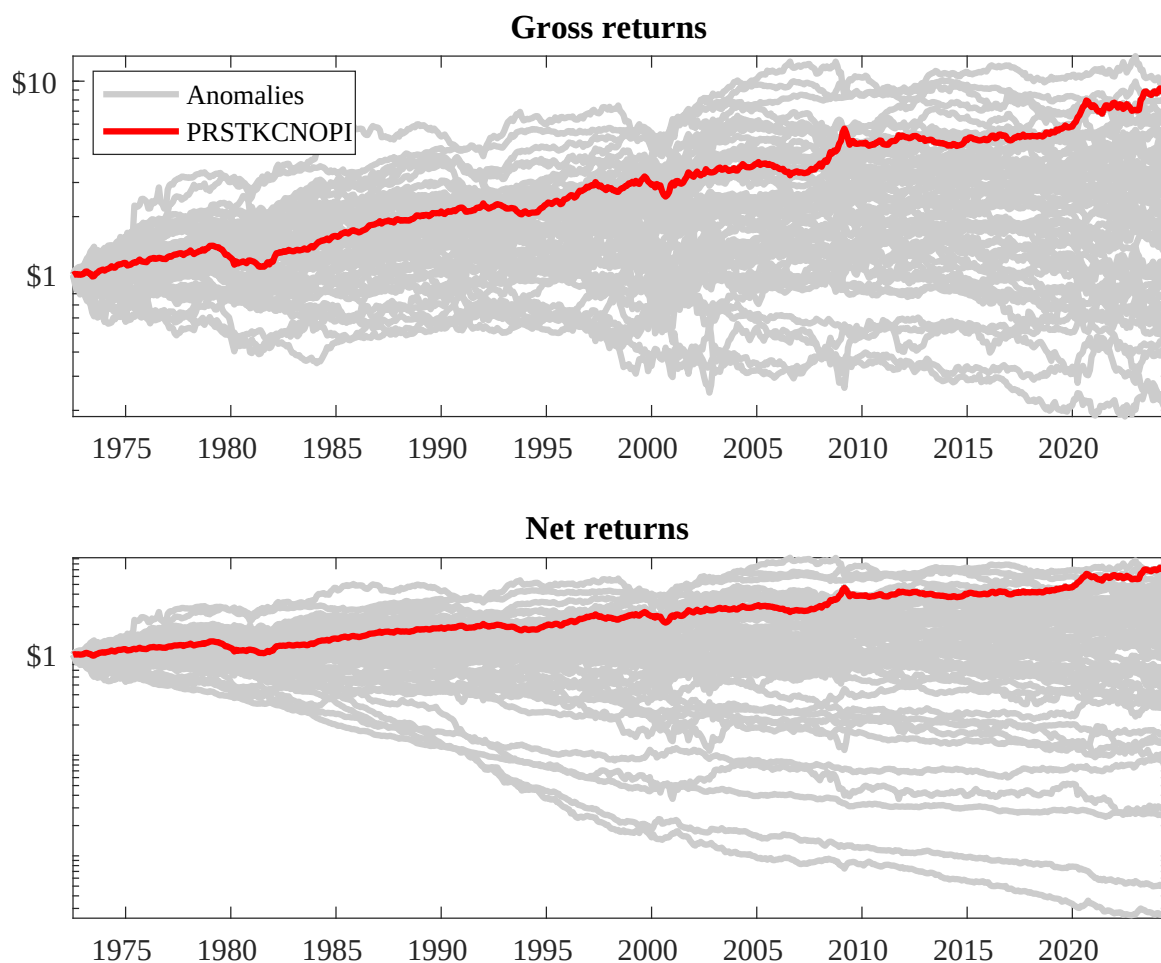
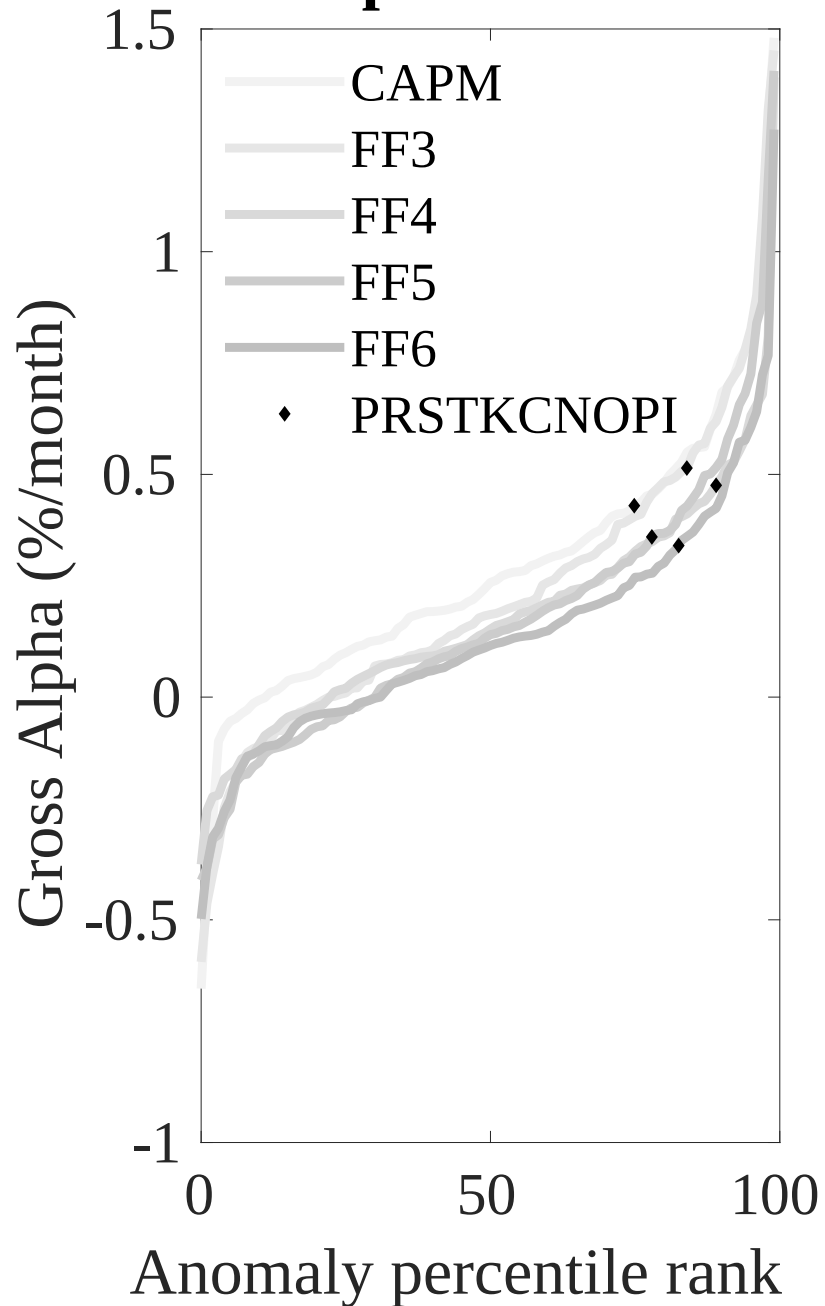


Figure 3: Dollar invested.

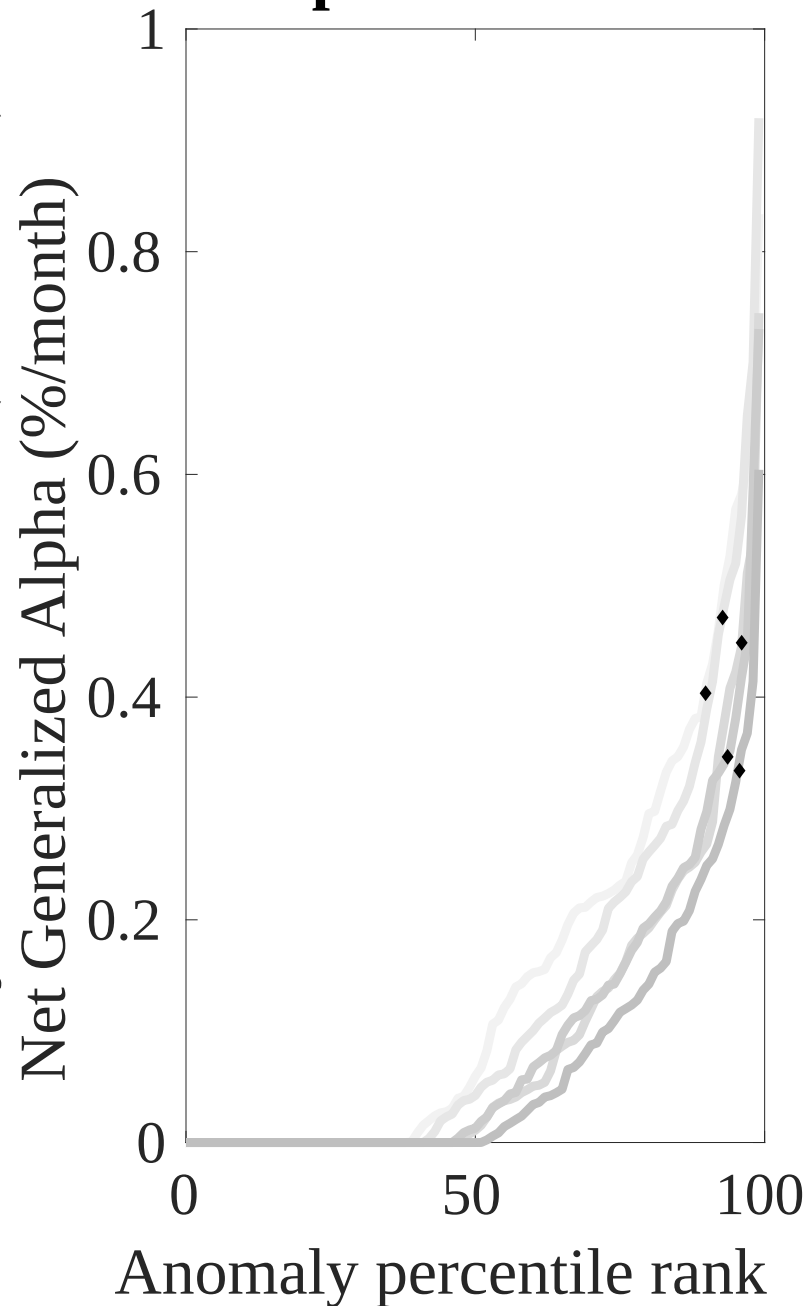
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the NBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



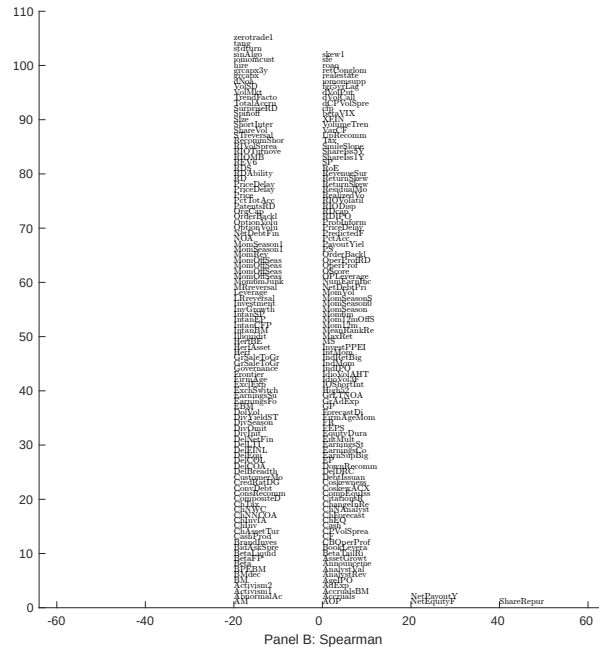
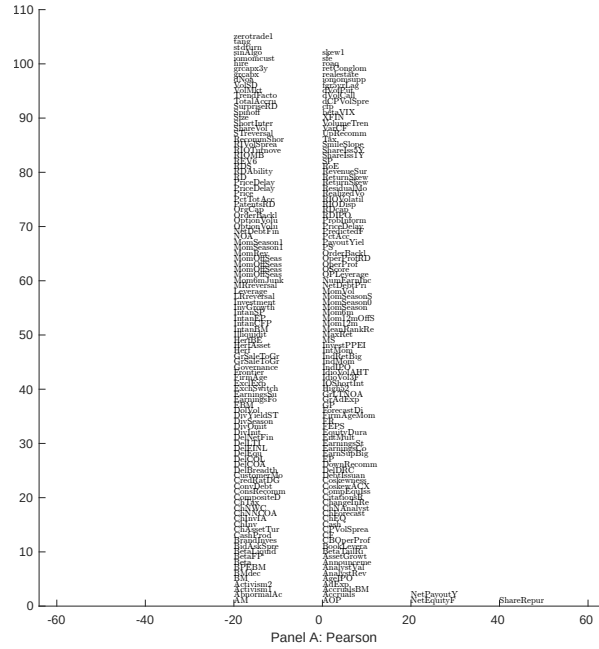
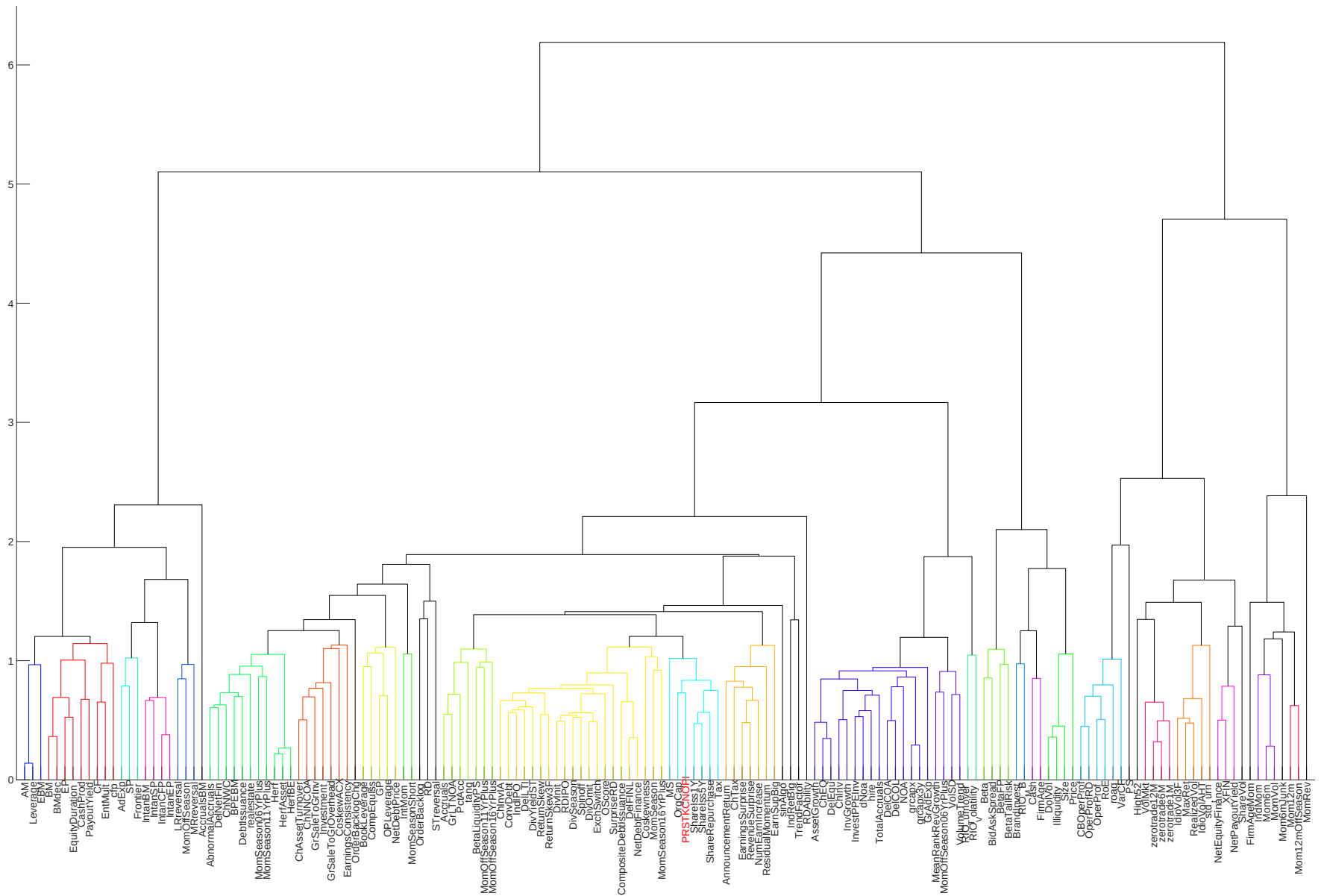


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with NBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



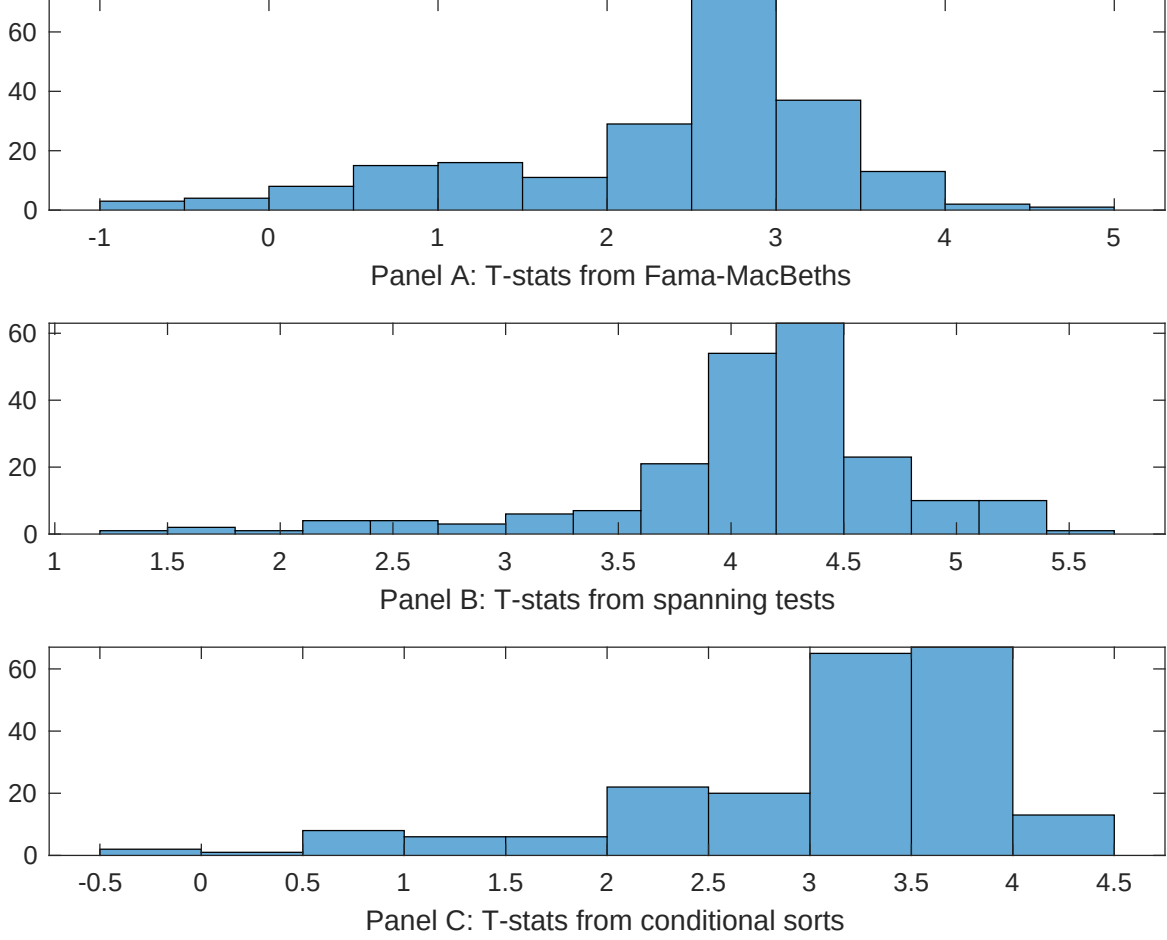


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of NBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NBR}NBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on NBR. Stocks are finally grouped into five NBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on NBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{NBR}NBR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.12 [5.28]	0.11 [3.74]	0.11 [4.62]	0.10 [3.93]	0.11 [4.57]	0.99 [3.45]	0.11 [3.88]
NBR	0.66 [1.53]	0.99 [2.03]	0.12 [2.75]	0.10 [2.26]	0.16 [2.32]	0.15 [3.29]	0.15 [2.65]
Anomaly 1	0.19 [7.16]						0.90 [3.44]
Anomaly 2		0.12 [2.96]					-0.35 [-0.77]
Anomaly 3			-0.26 [-0.23]				-0.11 [-0.86]
Anomaly 4				0.16 [4.79]			0.11 [3.02]
Anomaly 5					0.65 [3.07]		0.40 [2.74]
Anomaly 6						0.62 [1.14]	-0.37 [-0.09]
# months	624	619	624	619	616	576	576
$\bar{R}^2(\%)$	1	1	0	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the NBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{NBR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015a\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.32 [4.14]	0.29 [3.68]	0.34 [4.40]	0.30 [3.82]	0.35 [4.45]	0.34 [4.17]	0.27 [3.36]
Anomaly 1	24.50 [6.38]						23.34 [5.25]
Anomaly 2		14.37 [4.32]					9.33 [1.76]
Anomaly 3			13.55 [3.07]				7.04 [1.36]
Anomaly 4				12.00 [3.32]			0.13 [0.02]
Anomaly 5					2.78 [0.86]		-9.16 [-2.32]
Anomaly 6						6.81 [2.11]	0.68 [0.19]
mkt	-1.32 [-0.71]	-2.33 [-1.23]	-2.77 [-1.43]	-3.75 [-2.03]	-4.65 [-2.50]	-3.76 [-1.85]	0.16 [0.08]
smb	-4.08 [-1.40]	-7.18 [-2.45]	-7.40 [-2.39]	-8.24 [-2.79]	-11.64 [-4.16]	-12.10 [-3.34]	-4.81 [-1.29]
hml	-27.84 [-8.15]	-25.72 [-7.14]	-29.04 [-8.28]	-26.89 [-7.44]	-29.36 [-7.97]	-34.64 [-9.28]	-29.89 [-7.33]
rmw	12.80 [3.08]	16.93 [3.99]	15.32 [2.87]	21.69 [5.56]	24.71 [5.32]	20.99 [4.25]	9.61 [1.61]
cma	8.70 [1.53]	25.27 [4.83]	27.13 [5.06]	23.24 [4.40]	24.17 [4.54]	22.99 [4.20]	10.47 [1.70]
umd	2.86 [1.63]	1.59 [0.88]	2.93 [1.63]	1.99 [1.10]	2.48 [1.28]	2.39 [1.19]	4.58 [2.23]
# months	624	620	624	620	620	577	577
$\bar{R}^2(\%)$	32	30	28	29	27	32	38

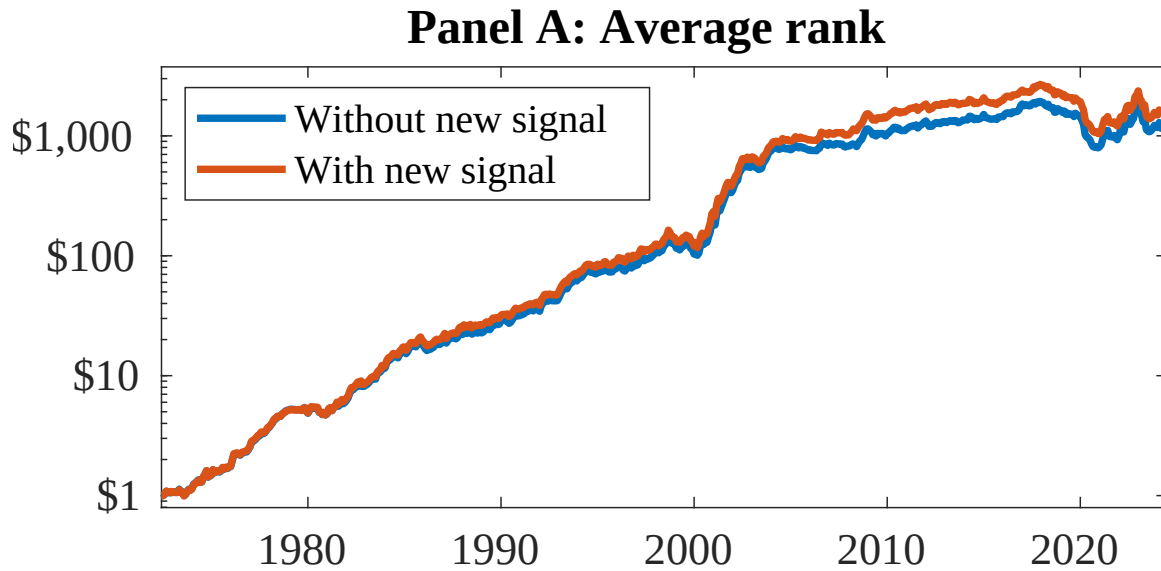


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as NBR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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